

## **Introduction**

Epilepsy is a neurological disorder characterized by chronic seizures from abnormal electrical activity in the brain, with a diverse range of causes and types (National Institute of Neurological Disorders and Stroke, 2025). Epilepsy is also one of the most prevalent neurological disorders in the United States, with over one million people affected by it in 2021 (Ney et al., 2025). Types of epilepsy can vary drastically in terms of clinical expression and severity, only linked together by the presence of chronic epileptic seizures, and often categorized by where seizures are localized in the brain (National Institute of Neurological Disorders and Stroke, 2025). Due to this variety, proper diagnosis is essential for choosing the correct treatment, of which there are also a great variety of (Santulli et al., 2016). The diagnosis of neurological disorders is very complex and can produce large amounts of data from procedures such as magnetic resonance imaging (MRI), electroencephalography (EEG), or even from a simple medical history (Shyam et al., 2026). Disorders caused by structural problems, such as brain tumors, also benefit from the use of immunohistochemical and pathological analysis to be diagnosed correctly (Thom et al., 2011). Due to the complexity of diagnosis and the amount of data available, artificial intelligence and deep learning technologies show potential in assisting the diagnosis of different types of epilepsy (Shyam et al., 2026).

## **Methods**

In order to correctly build a machine learning model to predict diagnosis from the available information, we must first understand the nature of the dataset we are training on. The dataset contains 298 records and contains 5 categories: MRI, immunohistochemistry, localization, microscopy and a diagnosis. Notably, some sections contained absent information. Next, we separated the dataset into one that contains the diagnosis, the item we were attempting to predict, and one that contains all other available medical information. The medical information contained 4 unique sections, so we delved deeper into the data by plotting the

distribution of frequency of occurrence of each word per section to learn more about the corpus we're working with, as can be seen in Figures 1 and 2.

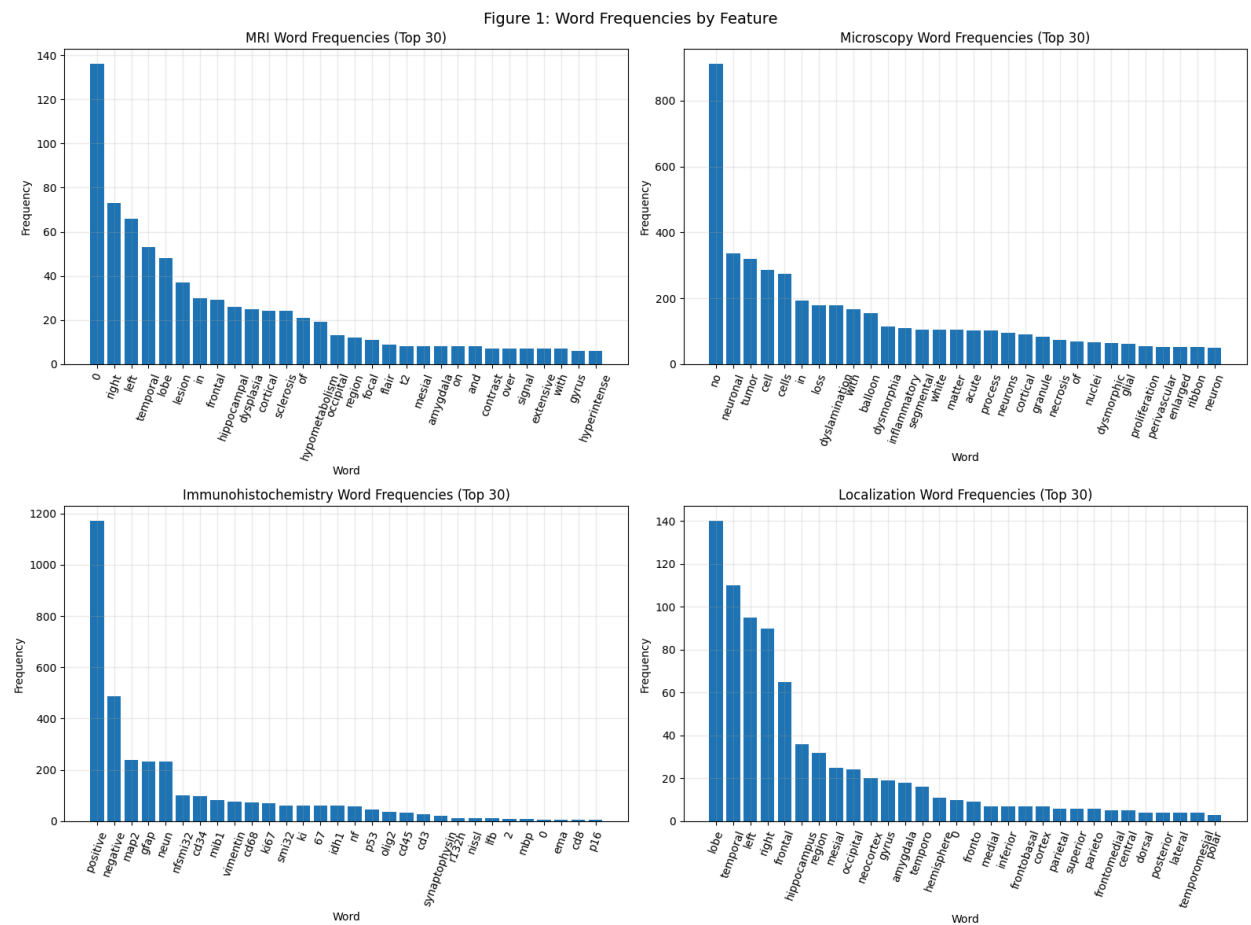


Figure 1: Word Frequency by Feature

Figure 2: Word Frequency by Rank

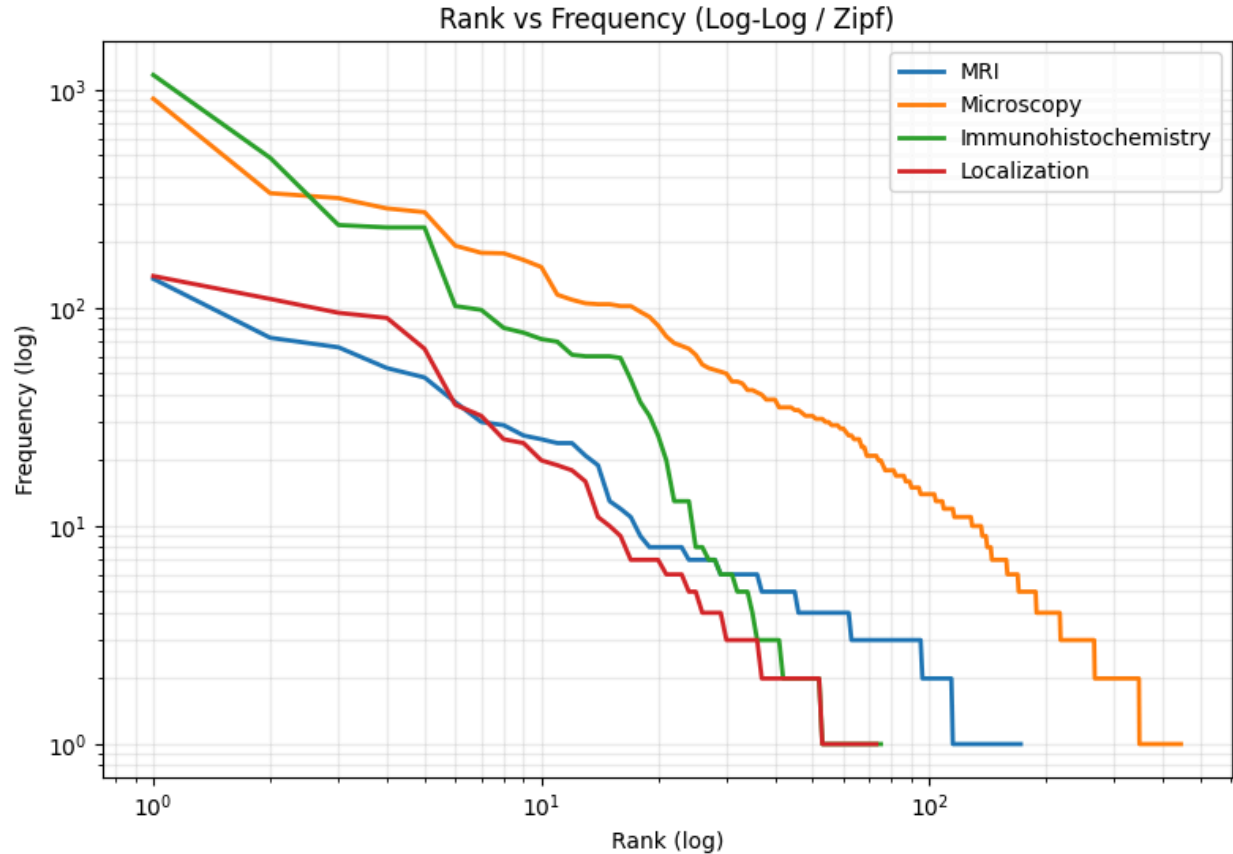


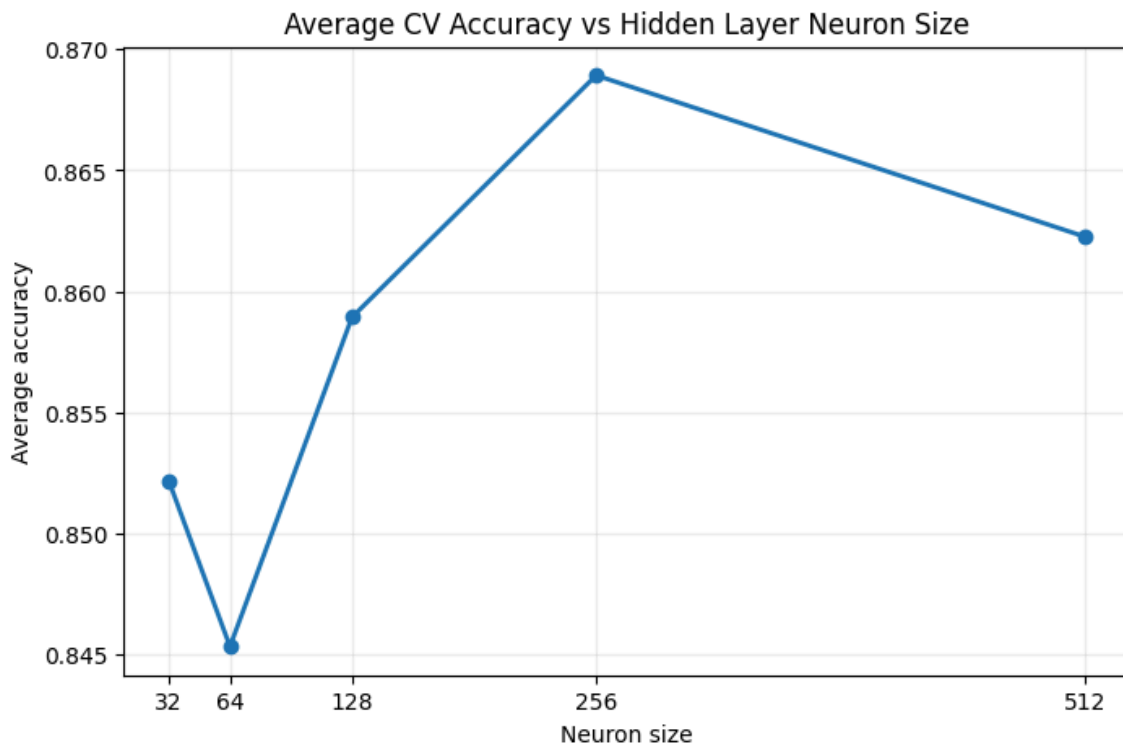
Figure 2: Rank-frequency Distribution

Each unique diagnosis was encoded with a number, with a total of 12 different diagnosis labels. The classes are significantly unbalanced and the dataset was quite small for training a neural network, so a stratified k-fold method was used to create 5 training and testing sets. In order to maintain compatibility between models the random seed was always kept consistent. For each fold approximately 80% of the data was used to train the model and 20% of the data was used to test the model. Once the dataset was split into its respective folds, a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer was created inside each fold's training set, respective to the feature category. Once vectorized, features across MRI, immunohistochemistry, localization, and microscopy were stacked to create our input features.

These stacked features and the corresponding encoded label were fed to a 3 layer densely connected sequential neural network. The number of neurons in each layer varied where the first layer had 32, 64, 128, 256, and 512 neurons and the second layer had half the number as the first layer. Both layers used the ReLU activation function, and the output layer had a softmax activation function with 12 neurons, 1 for each class. While training, 10% of the training set was used as validation. After fitting the model, predictions were derived from the testing set. The training and validation losses were graphed alongside a confusion matrix.

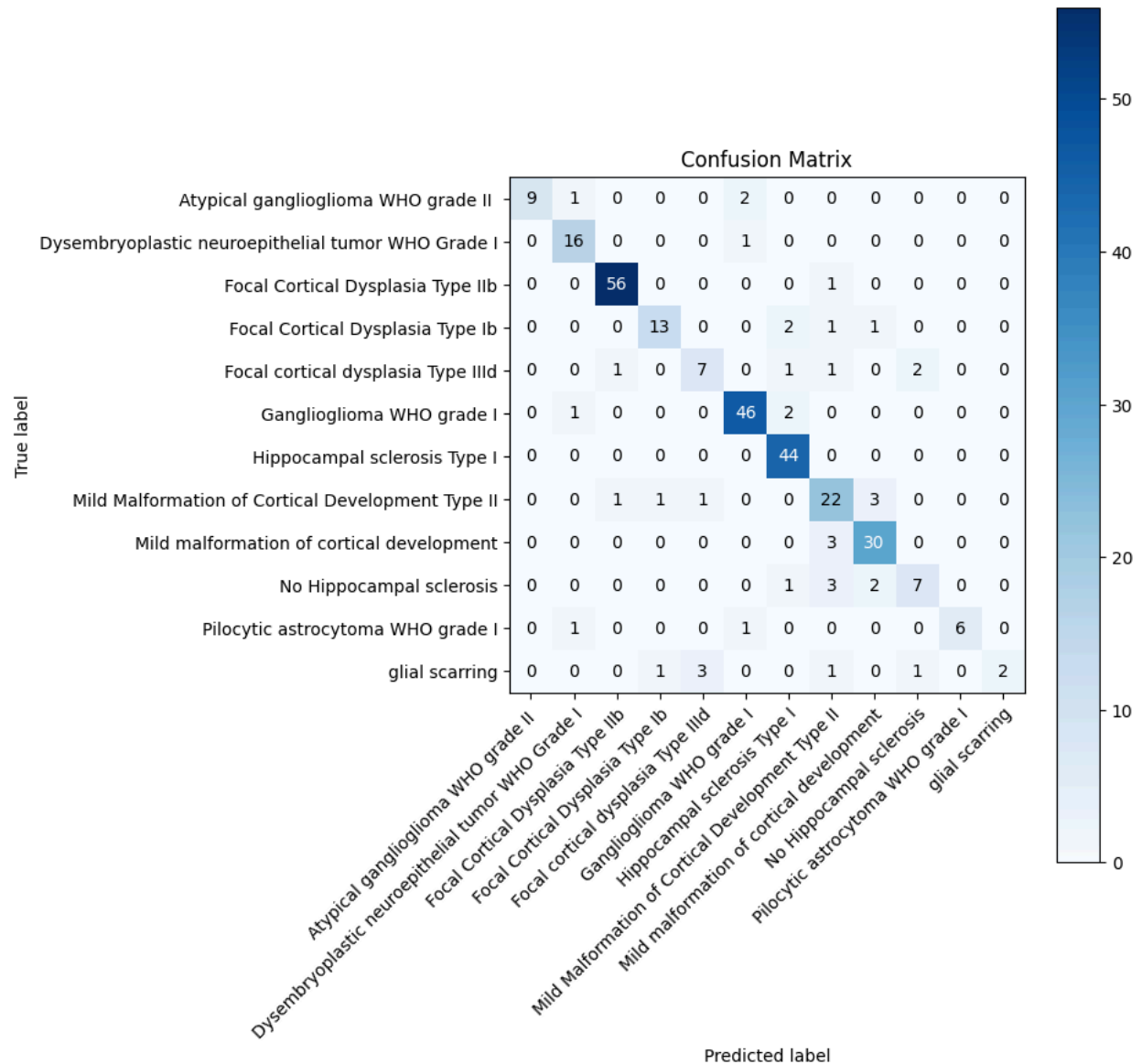
## Results

Classification accuracy with a TFIDF-DNN classification consistently measured around 85% percent accuracy. Modifying the neuron sizes of the input and hidden layer did not significantly improve the accuracy of the model, as can be seen in Figure 3.



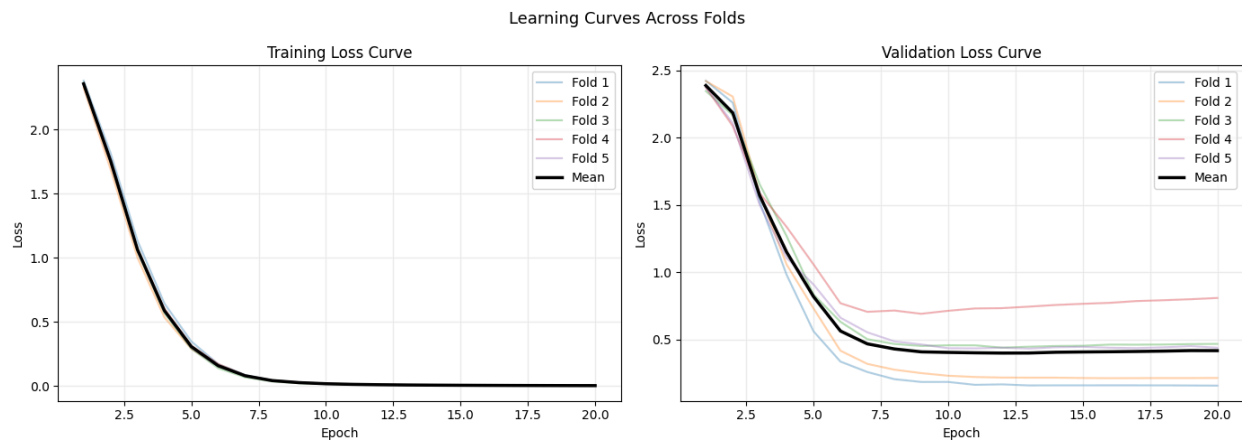
**Figure 3:** Accuracy by neuron size

The confusion matrix shown in Figure 4 shows how often each diagnosis is correctly predicted. No diagnosis is predicted incorrectly significantly more often than the others and even though the classes are very unbalanced, the neural network predicts all diagnoses at least once. Figure 4 demonstrates a model with 64 neurons and has a final accuracy of 86.58%.



**Figure 4:** Confusion Matrix. Neuron Size = 64

Training and validation loss curves for neuron size 64, shown in Figure 5, look healthy and show a decrease in both training and validation loss over epochs on average.



**Figure 5:** Training and Validation Loss. Neuron Size = 64

## Discussion

Our results show that TF-IDF DNN classifier achieves approximately 85% classification accuracy on the provided dataset. This significantly outperformed our previous attempts with a more complicated RNN with a BERT embedding layer and a late-fusion neural network. It is probable the TF-IDF DNN classifier performed well due to the type of text contained in the dataset. Figure 1 shows a significant portion of each category's features were composed of a few words, while Figure 2 shows the dataset does not show a word's rank inversely proportional to its frequency, as expected in natural language. This supports the idea the dataset's verbiage contains shorter, jargon-laden medical terminology lacking a typical sentence structure. It's likely the differences between categories rely on a few high-impact words instead of sentence-based logic, which made TF-IDF optimal. It is noteworthy that the dataset was also very small and it is questionable if TF-IDF DNN could generalize to a much larger dataset contributed to by multiple different systems. It is also likely the model's performance would degrade if the records

contained many synonyms, diverse verbiage, or complex structured sentences. These are areas a more sophisticated model could outperform TF-IDF DNN.

Increasing the neuron size did not meaningfully increase the performance of the model. It is likely that the minor fluctuations in the performance between neuron sizes is due to the stochastic nature of the weight initialization and k-fold stratification. It's possible this can be attributed to the small size of the dataset, or a relatively simple underlying manifold. Figure 4 showed errors were generally random and evenly distributed between classes, even though the classes were unbalanced. Even the glial scarring label, which was only represented 8 times in the data, was predicted twice, demonstrating the random nature of errors. The validation and training loss curves were healthy. Although, it is important to notice that in Figure 5 that not all folds consistently decrease. For example, fold 4 actually increases past the 8th epoch and is likely overtraining. A future model could include a patience parameter to stop overtraining on specific folds by stopping early.

## References

National Institute of Neurological Disorders and Stroke. (2025, April 7). *Epilepsy and Seizures*.

National Institutes of Health.

<https://www.ninds.nih.gov/health-information/disorders/epilepsy-and-seizures>

Ney, J. P., Steinmetz, J. D., Anderson-benge, E., Gillespie, C. W., Becker, A., Steele, X., &

Esper, G. J. (2025). US Burden of Disorders Affecting the Nervous System. *JAMA*

*Neurology*, 83(1). <https://doi.org/10.1001/jamaneurol.2025.4470>

Santulli, L., Coppola, A., Balestrini, S., & Striano, S. (2016). The challenges of treating epilepsy with 25 antiepileptic drugs. *Pharmacological Research*, 107, 211–219.

<https://doi.org/10.1016/j.phrs.2016.03.016>

Shyam, M., Kaushal, N., Yadav, M., Srirangan, P., & Prince, S. E. (2026). Transforming

Epilepsy Care: A Comprehensive Review of Technological Innovations, Artificial

Intelligence Advancements, and Precision Tools for Enhanced Seizure Management and

Patient Outcomes. *Scientifica*, 2026(1). <https://doi.org/10.1155/sci5/5915062>

Thom, M., Toma, A. K., An, S., Martinian, L., Hadjivassiliou, G., Ratilal, B., Dean, A., McEvoy, A.

W., Sisodiya, S. M., & Brandner, S. (2011). One Hundred and One Dysembryoplastic

Neuroepithelial Tumors: An Adult Epilepsy Series With Immunohistochemical, Molecular

Genetic, and Clinical Correlations and a Review of the Literature. 70(10), 859–878.

<https://doi.org/10.1097/nen.0b013e3182302475>